



MMA093 Rigid Body Dynamics

Assignment 2

Mandatory Part

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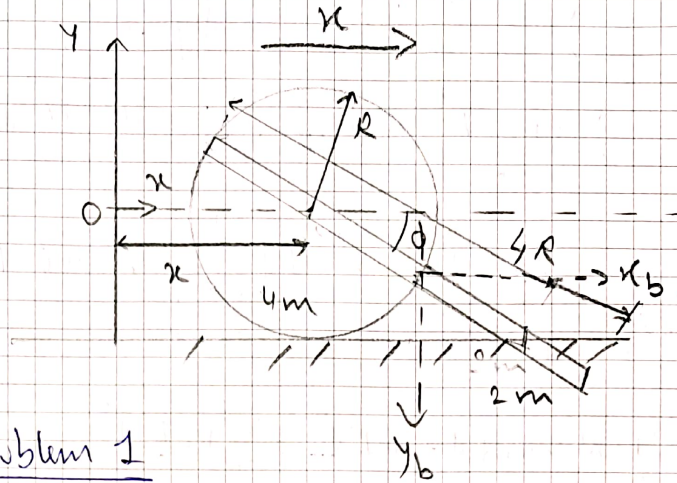
Chalmers University of Technology

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Problem 1: Equation of motion

The Derivation of Equation of motion of the system



Problem 1

$$m = 1 \text{ kg} \quad R = 0.1 \text{ m}$$

$$M_c = 4m = 4 \text{ kg}$$

$$M_b = 2m = 2 \text{ kg}$$

$$L = 4R = 0.4 \text{ m}$$

$$\omega = \frac{x}{R} = \dot{\omega} = \frac{\dot{x}}{R}$$

$$I_c = \frac{1}{2} M_c R^2$$

$$I_b = \frac{1}{12} M L^2$$

KE of cylinder

$$\text{Translation KE}_{\text{cyl}} = \frac{1}{2} m_c v^2 = \frac{1}{2} M_c \dot{x}^2 = 2m \dot{x}^2$$

$$\text{Rot KE}_{\text{cyl}} = \frac{1}{2} I \dot{\omega}^2 = \frac{1}{2} 2mR^2 \frac{\dot{x}^2}{R^2} = m \dot{x}^2$$

$$\begin{aligned} \text{Total KE}_{\text{cyl}} &= \text{Translation KE}_{\text{cyl}} + \text{Rot KE}_{\text{cyl}} \\ &= 2m \dot{x}^2 + m \dot{x}^2 = 3m \dot{x}^2 \quad \text{--- (1)} \end{aligned}$$

KE of Bar

$$\text{No slipping } \dot{x}_c = R \dot{\theta}$$

$$y_b = \begin{bmatrix} R \cos \phi + x \\ -R \sin \phi \end{bmatrix}$$

$$\dot{y}_b = \begin{bmatrix} R(-\sin \phi) \cdot \dot{\phi} + \dot{x} \\ -R \cos \phi \cdot \dot{\phi} \end{bmatrix}$$

$$\text{Kinetic Energy of Bar} = \frac{1}{2} \times 2m\dot{x} \left[\dot{x} - R \sin\phi \dot{\phi} \right]^2 + \frac{1}{2} \times 2m \times R^2 \cos^2\phi \cdot \dot{\phi}^2 + \frac{1}{2} \times \frac{2m(16R^2)}{12} \dot{\phi}^2$$

$$\Rightarrow m \left[\dot{x}^2 - 2\dot{x}R \sin\phi \dot{\phi} + R^2 \sin^2\phi \dot{\phi}^2 \right] + mR^2 \cos^2\phi \cdot \dot{\phi}^2 + \frac{4}{3} mR^2 \dot{\phi}^2$$

$$\Rightarrow m\dot{x}^2 - 2m\dot{x}R \sin\phi \dot{\phi} + mR^2 \dot{\phi}^2 + \frac{4}{3} mR^2 \dot{\phi}^2 - \frac{4}{3} mR^2 \dot{\phi}^2 - 2m\dot{x}R \sin\phi \dot{\phi} + m\dot{x}^2 \quad \text{--- (2)}$$

Total kinetic Energy Eq (1) + Eq (2).

$$= \frac{4}{3} mR^2 \dot{\phi}^2 - 2m\dot{x}R \sin\phi \dot{\phi} + 4m\dot{x}^2 \quad \text{--- (3)}$$

Potential Energy V.

$$V = -2mgR \sin\phi \quad \text{--- (4)}$$

Lagrangian L = T - V Eq (3) - (4).

$$L = \frac{4}{3} mR^2 \dot{\phi}^2 - 2m\dot{x}R \sin\phi \dot{\phi} + 4m\dot{x}^2 + 2mgR \sin\phi$$

For x.

$$\frac{\partial L}{\partial \dot{x}} = 0 \Rightarrow \frac{\partial}{\partial \dot{x}} \left(-2m\dot{x}R \sin\phi \dot{\phi} + 4m\dot{x}^2 \right)$$

$$\Rightarrow 8m\dot{x} - 2mR \sin\phi \dot{\phi}$$

$$\frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{x}} \right) = -2mR \left[\dot{\phi} \cos \phi \cdot \dot{\phi} + \sin \phi \cdot \ddot{\phi} \right] + 8m\ddot{x}$$

$$= 8m\ddot{x} - 2mR \sin \phi \ddot{\phi} - 2mR \cos \phi \cdot \dot{\phi}^2$$

$$\text{For } \phi = \left[8m\ddot{x} - 2mR (\sin \phi \ddot{\phi} + \cos \phi \dot{\phi}^2) = 0 \right] \quad (5)$$

For ϕ

$$\frac{\partial L}{\partial \dot{\phi}} = 0 \Rightarrow \frac{\partial}{\partial \dot{\phi}} \left(\frac{\mp}{3} mR^2 \dot{\phi}^2 - 2m\dot{x}R \sin \phi \dot{\phi} \right)$$

$$\Rightarrow \frac{14}{3} mR^2 \ddot{\phi} - 2m\dot{x}R \sin \phi$$

$$\frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = \frac{14}{3} mR^2 \ddot{\phi} - 2mR (\ddot{x} \sin \phi + \dot{x} \cos \phi \cdot \dot{\phi})$$

$$= \frac{14}{3} mR^2 \ddot{\phi} - 2mR \ddot{x} \sin \phi - 2\dot{x} \cos \phi \cdot \dot{\phi} \cdot mR$$

Eqn

$$= \frac{14}{3} mR^2 \ddot{\phi} - 2mR \ddot{x} \sin \phi - 2\dot{x} \cos \phi \dot{\phi} \cdot mR + 2m\dot{x}R \cos \phi \dot{\phi}$$

$$- 2mgR \cos \phi = 0$$

$$\Rightarrow \left[\frac{\mp}{3} R \ddot{\phi} - \ddot{x} \sin \phi - g \cos \phi = 0 \right] \quad (6)$$

Here x is a cyclic coordinate.

Problem 2: Angular Velocity of the Bar

Problem 2

Initially rest condition $\kappa=0$ $\phi=0$ $\dot{\phi}=0$.

from Eq (5) problem 1.

$$8m\ddot{\kappa} - 2mR\sin\phi\dot{\phi} = 0.$$

Divide by $4m$ $2\ddot{\kappa} - R\sin\phi\dot{\phi} = 0$.

$$\boxed{\ddot{\kappa} = \frac{R\sin\phi\dot{\phi}}{4}}.$$

$$KE_{\text{initial}} = 0 \quad PE_{\text{initial}} = 0.$$

$$T_i + V_i = 0$$

from Eq (3) & (4) from problem 1.

$$T_j = \frac{7}{3}mR^2\dot{\phi}^2 - 2m\dot{\kappa}R\sin\phi\dot{\phi} + 4m\dot{\kappa}^2$$

$$V_j = -2mgR\sin\phi.$$

Bar is vertical $\phi = \frac{\pi}{2}$.

$$T_j = \frac{7}{3}mR^2\dot{\phi}^2 - 2m\dot{\kappa}R\dot{\phi} + 4m\dot{\kappa}^2$$

$$V_j = -2mgR.$$

$$T_i + V_i = T_j + V_j$$

$$= \frac{7}{3}mR^2\dot{\phi}^2 - 2m\dot{\kappa}R\dot{\phi} + 4m\dot{\kappa}^2 - 2mgR. \quad - (7)$$

Substitute $\ddot{\kappa}$ in Eq (7).

$$\Rightarrow 4m\left[\frac{1}{4}R\sin\phi\dot{\phi}\right]^2 + \frac{7}{3}mR^2\dot{\phi}^2 - 2m\left[\frac{1}{4}R\sin\phi\dot{\phi}\right]R\dot{\phi} - 2mgR.$$

$$\Rightarrow 4m \left(\frac{1}{16} R^2 \sin^2 \phi \dot{\phi}^2 \right) + \frac{7}{3} m R^2 \dot{\phi}^2 - 2m R^2 \phi^2 \frac{1}{4} \sin \phi - 2mgR = c$$

$$\dot{\phi}^2 = \frac{24g}{25R} \Rightarrow \dot{\phi} = \sqrt{\frac{24g}{25R}}$$

$$\dot{\phi} = \sqrt{\frac{24 \times 9.81}{25 \times 0.1}} = 9.704 \text{ rad/sec}$$

$$\boxed{\dot{\phi} = 9.704 \text{ rad/s}}$$

Problem 3. ADAMS

The model in the ADAMS is created with the specified paramters.

The contact use with same static and dynamic coefficient is condiseder as 1 which approximates no slip and stiction and friction transition velocities is set to very small value. The angle obtained in ADAMS is -1.57 rad.

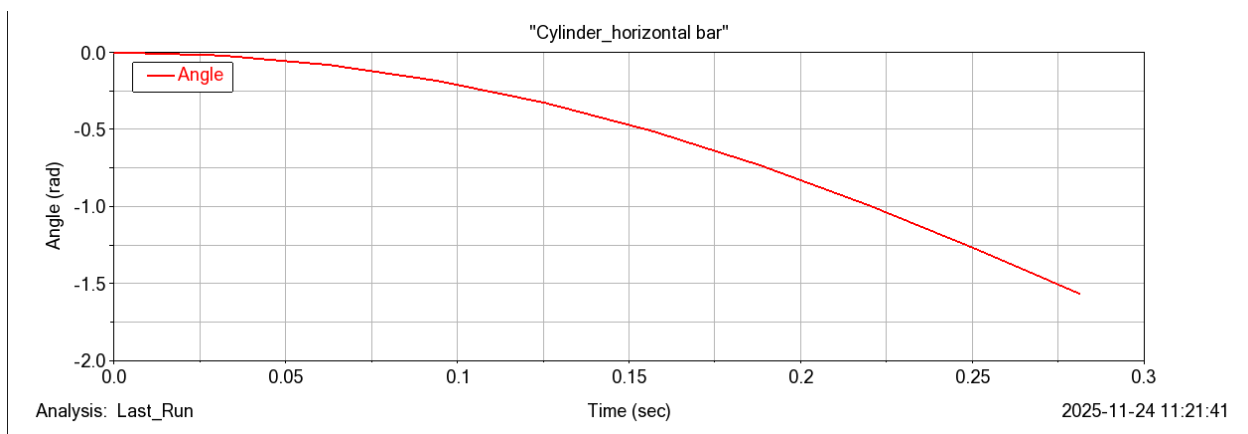


Figure 1: Angle Measure(rad)



Figure 2: Angular Velocity(rad/s)

The angular velocity obtained analytically is 9.70 rad/sec and angular velocity obtained in ADAMS is 9.69 rad/sec. As we can see the angular velocity obtained through both the methods are very close.